

Semirings for database provenance

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What is a database?

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Loans

Loan_id	Member_id	Book_id	Borrow_date	Duration
43666	4221	389	17-02-2026	7
43666	4221	782	17-02-2026	7
43952	4222	975	17-02-2026	30
54673	4223	389	06-03-2026	14
58930	4224	975	09-03-2026	30
58930	4224	879	09-03-2026	30

Books

Book_id	Book_title	Genre	Copies
389	Nineteen Eighty-Four	Fiction	5
782	Brave New World	Fiction	4
879	Divine Comedy	Poetry	6
975	Critique of Pure Reason	Philosophy	3

Members

Member_id	First_name	Last_name
4221	Aditya	Prakash
4222	Graham	Leigh
4223	Ivan	Di Liberti
4224	Luigi	Brunetta

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- A *database* D as a finite set of relations.

How do we query a database?

A query specifies how new relations can be derived from an existing database. Relational algebra is an algebraic formalism in which queries are expressed by applying a sequence of operations to relations.

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The *relational algebra* consists of the following operations:

- Selection, $\sigma_{\theta}(R)$: Keep tuples satisfying condition θ
- Projection, $\pi_L(R)$: Keep only attributes L
- Natural join, $R \bowtie S$: Merge tuples matching on common attributes
- Renaming, $\rho_{\beta}(R)$: Rename attributes according to β
- Union, $R \cup S$: Combine tuples present in R or in S
- Difference, $R - S$: Keep tuples present in R but not in S

Example

Query: Find the title of any book borrowed for 7 or 14 days.

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In relational algebra we can express it with the following expression:

$$\pi_{Book_title}(\sigma_{Duration=7}(Loans \bowtie Books) \cup \sigma_{Duration=14}(Loans \bowtie Books))$$

What is database provenance?

Provenance is a mechanism that relates data in an output to the contributing data in the input.

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It is typically represented as a form of annotation information, usually by tagging data on the tuple-level of a relation.

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43666	4221	782	17-02-2026	7	l_2
43952	4222	975	17-02-2026	30	l_3
54673	4223	389	06-03-2026	14	l_4
58930	4224	975	09-03-2026	30	l_5
58930	4224	879	09-03-2026	30	l_6

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389	Nineteen Eighty-Four	Fiction	5	b_1
782	Brave New World	Fiction	4	b_2
879	Divine Comedy	Poetry	6	b_3
975	Critique of Pure Reason	Philosophy	3	b_4

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4222	Graham	Leigh	m_2
4223	Ivan	Di Liberti	m_3
4224	Luigi	Brunetta	m_4

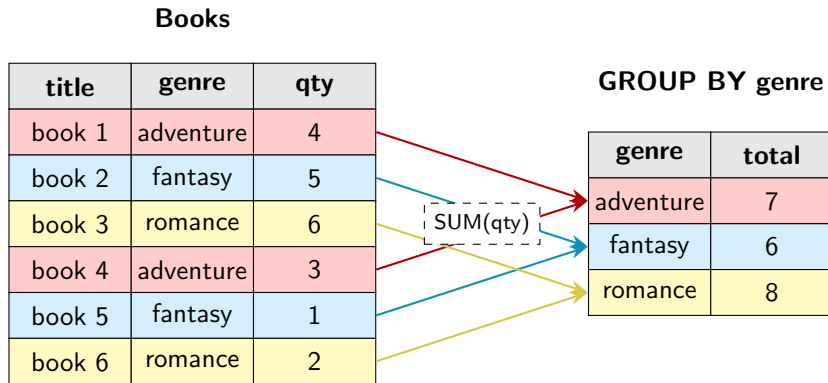
What is a semiring?

Semiring framework

Extending the semiring framework

Extensive work has been carried out to extend the semiring framework to more expressive

Aggregate queries



Aggregation provenance

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- Unified framework for recursion, aggregates, and difference
- Trade-offs between generality and usability
- Limits of semiring-based provenance